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ORIGINAL PAPER

# Multi-party reversible data hiding in shared medical images based on texture-guided hierarchical quantization coding

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## Abstract
In recent years, the advancement of smart healthcare has heightened concerns over medical information privacy. Reversible data hiding technology has been applied to medical images to ensure both images and patient data remain secure and tamper-free. In this paper, we propose a novel multi-party reversible data hiding scheme based on texture-guided hierarchical quantization coding for medical information security and privacy. The core of our approach is a novel Texture-Guided Hierarchical Quantization Coding (TGHQC) method, which can leverage the texture features of pixels to guide the hierarchical quantization coding process, thereby increasing the vacated room and enhancing embedding capacity. In addition, based on TGHQC, we further propose a multiple embedding method that not only supports embedding various types of data but also maximizes embedding efficiency while ensuring information security. Finally, we present a hierarchical sharing multi-party management framework that integrates TGHQC with the new embedding method, allowing hospitals to access patient privacy information and medical images only with explicit patient authorization. Our multi-party reversible hiding method effectively safeguards patient privacy and ensures secure storage and access to information within medical systems. Experimental results demonstrate the scheme's superior embedding capacity and lossless reconstruction.

**Keywords** Healthcare privacy · Medical image · Reversible data hiding · Secret image sharing · Hierarchical quantization encoding

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## 1 Introduction

With the rapid development of multimedia and internet technology, and as human society swiftly advances toward an information society, people's lives are increasingly integrated with the internet, driving transformative changes. In this information age, the healthcare industry has been profoundly impacted, with automated healthcare technologies progressing rapidly, giving rise to telemedicine and various medical management systems. Medical institutions are continuously establishing and refining medical information management systems, with various types of medical data gradually becoming digitized. Digital medical management systems have significantly improved the efficiency of managing, storing, and transmitting patient information, saving time and cost. However, this advancement also brings new challenges: medical information transmitted over networks is vulnerable to interception and tampering by malicious actors, leading to potential breaches of patient privacy and even misdiagnosis. Additionally, the critical information embedded within medical images transmitted in telemedicine must be fully protected, while image quality must be maintained to enable accurate diagnosis. Therefore, the transmission of medical images in telemedicine presents numerous challenges (Huang et al. 2024).

Reversible Data Hiding (RDH), also known as lossless data hiding or reversible information hiding, is an innovative technology that allows secret information to be embedded in a plaintext carrier while ensuring that the carrier data can be perfectly restored once the embedded information is extracted. This technology is increasingly used with multimedia data, such as images, audio, and video. Common reversible data hiding algorithms include Difference Expansion (DE) (Tian 2003), Histogram Shifting (HS) (Ni et al. 2006; Thodi and Rodríguez 2007), Prediction Error (PE)-based methods (Li et al. 2013), and Pixel Value Ordering (PVO)-based methods (Luo et al. 2010). These methods aim to maximize embedding capacity while preserving image quality, and they are widely applied in scenarios such as copyright protection, data management, and information verification for multimedia content.

However, medical images inherently carry a wealth of sensitive patient information, making privacy protection crucial. In addition, medical images come with various auxiliary information, such as patient identity, medical records, and diagnostic details. In this context, Reversible Data Hiding in Encrypted Images (RDHEI) (Zhang 2011; Huang et al. 2016; Zhang 2012; Ma et al. 2013; Yin et al. 2022; Yu et al. 2022; Yin et al. 2022; Yao et al. 2024; Yu et al. 2022; Yin et al. 2020) has emerged as a more suitable technology for transmitting and storing medical images. RDHEI was first proposed by Zhang (2011). They encrypted an original image of size $m \times n$ by performing a difference operation with a random key, then divided the encrypted image into blocks of $8 \times 8$ and embedded one pixel in each block, achieving a data embedding capacity of $\frac{m \times n}{64}$. A random $m \times n$ 01 matrix was generated and divided into two sets: $S_0$, containing 0 s, and $S_1$, containing 1 s. If the data to be embedded in each block was 0, the last three bits of $S_0$ were flipped according to the bit, while $S_1$ remained unchanged, and vice versa.

To improve data extraction accuracy, some modifications were introduced in subsequent research (Huang et al. 2016), though these methods lacked separability between image recovery and data extraction. In response, Zhang et al. proposed a separable approach (Zhang 2012) that further optimized embedding space. Subsequently, other methods were proposed, with one of the most representative approaches using a Median Edge Detector (MED) for pixel prediction (Ma et al. 2013). This approach calculates the difference between the predicted and original pixel values, embedding data from the most to the least significant bit. Additionally, auxiliary data to be embedded is Huffman encoded to enhance efficiency.

To further enhance image transmission security, scientists have proposed combining Reversible Data Hiding (RDH) with Secret Sharing (SS), leading to the RDHSI method (Wu et al. 2018; Qin et al. 2021; Xiong et al. 2023; Weng et al. 2023; Chen et al. 2022; Yang et al. 2007; Chen et al. 2019). This combination not only encrypts the image but also enables recovery in case of transmission errors. In Wu et al. (2018), the authors first proposed using Shamir's Secret Sharing (SS) (Thien and Lin 2002; Shamir 1979) for reversible information hiding, adopting methods such as difference expansion and histogram shifting. Since Shamir's SS method cannot effectively handle pixel values greater than 251, scientists suggest using Galois Field $GF(2^8)$ instead of $GF(251)$ (Qin et al. 2021), allowing shared images to be recovered without redundant information.

Multi-party Reversible Data Hiding (Chen et al. 2024) extends the RDHSI technique, enabling multiple participants to embed hidden information while sharing data, ensuring that the original data can be perfectly restored after extraction. Most current reversible information hiding methods based on secret image sharing focus on evenly dividing the entire image, overlooking the distinct priorities of different parties (Chen et al. 2020). However, for different roles, certain regions within an image may be of greater importance, and it may not always be necessary to extract the complete image. For instance, doctors may only need to access specific lesion areas, while hospitals may require comprehensive diagnostic data. Medical images often contain highly sensitive patient information, which requires both privacy protection and content integrity during distribution, storage, and sharing to facilitate subsequent diagnosis and treatment.

To address the privacy and data integrity risks of medical images in multi-party collaboration scenarios, this paper proposes a multi-party reversible data hiding scheme based on texture-guided hierarchical quantization coding. The scheme integrates hierarchical image sharing with a dual embedding mechanism, achieving high-capacity, secure management of medical images for multiple parties. Additionally, the scheme features reversibility and separability, enabling different roles to access data according to specific requirements. The main contributions of this paper are as follows:

1. A multi-party reversible data hiding framework for healthcare systems is proposed to facilitate secure storage and access to images. The framework assigns different access levels to authorized parties, enabling them to extract secret data and related information from specific regions of the image. This method eliminates additional storage requirements while ensuring secure access to secret data and images. By meeting the unique data needs of each party and protecting sensitive information, the scheme effectively safeguards patient privacy and data integrity.

2. A texture-guided hierarchical quantization coding method is proposed, which adaptively determines block types based on the texture features of pixel blocks and applies hierarchical quantization to difference information to reduce storage space. By dynamically adjusting coding strategies according to image content, this method optimizes embedding efficiency.

3. A dual embedding mechanism is designed, with the level 1 embedding performed by the image owner, specifically embedding medical image metadata. The level 2 embedding is carried out by data embedders, where patients and hospitals embed their respective information into corresponding parts of the image. This embedding method significantly increases embedding capacity while ensuring information security. Moreover, it allows the system to flexibly meet diverse data access requirements, thereby safeguarding data security and privacy.

The remainder of this paper is organized as follows: Section 2 introduces the relevant technologies of Secret Sharing (SS) and Reversible Data Hiding (RDH). Section 3 details the proposed multi-party reversible data hiding scheme based on texture-guided hierarchical quantization coding. Section 4 discusses the experimental results. Section 5 concludes the paper.

## 2 Related work

### 2.1 Secret sharing

Secret sharing was proposed by researchers (Thien and Lin 2002) to address common single point of failure issues and access control problems in key-based systems, preventing authorized personnel from disclosing secret information. This section will detail the basic concept of secret sharing to help readers understand the methodology used in this paper.

A secret sharing scheme involves a two-phase access structure for a secret $S$, with a set of $n$ authorized parties and a set of $t$ participants. The secret $S$ is divided into $n$ sub-shares, which are distributed to $n$ authorized parties for safekeeping. Any set of $t$ participants can access the secret $S$. The secret sub-shares held by each authorized party do not contain any information about the original secret $S$. Even if the sub-shares of $n - t$ authorized parties are compromised, the remaining $t$ sub-shares can still be used to reconstruct the secret, ensuring the recovery of $S$ is not affected. Secret sharing schemes utilize a two-phase access structure, effectively solving these issues and significantly enhancing the security of data privacy protection.

The polynomial-based $(t, n)$ secret sharing algorithm proposed by Shamir is fundamental in the field of secret sharing. The basic idea is to construct a set of polynomial equations over a finite field $F$, using the secret $S$ as the polynomial coefficients to generate multiple encrypted shares. Secret receivers can recover the secret $S$ by reconstructing the polynomial using some of the shares. This $(t, n)$ secret sharing algorithm has three main advantages:

1. At least $t$ shared shares are needed to recover the secret $S$;

2. No single share can reveal information about the secret $S$;

3. It is fault-tolerant, allowing up to $n - t$ shares to be lost or corrupted.

Thien and Lin extended Shamir's secret sharing scheme (Thien and Lin 2002) to the image domain by proposing a $(t, n)$ threshold secret image sharing algorithm. In this algorithm, the pixel values of an image are used as polynomial coefficients to generate shadow images, and at least $t$ shadow images can be used to recover the original secret image using Lagrange interpolation polynomials.

In this scheme, since some pixel values are truncated to 250 before sharing, the recovered image is not identical to the original image. Therefore, this algorithm is not suitable for applications that require high accuracy in the recovered images and has certain limitations. To address this, Thien and Lin (2002) proposed a complementary scheme in which all pixel values greater than or equal to 250 are shared by splitting them into two values, resulting in an increase in the size of the generated shadow images.

To solve the above problems, Yang et al. (2007) proposed using the Galois field $GF(2^8)$ instead of $GF(251)$, reconstructing the original polynomials as follows:

$$f_j(x) = p_{j,1} + p_{j,2}x + \ldots + p_{j,t}x^{t-1} \bmod g(x) \tag{1}$$

$$L_j(x) = \sum_{t=1}^{t} (f_j(x_i) \times \prod_{0<k<t, k\neq i} \frac{x - x_k}{x_i - x_k} \bmod g(x)) \tag{2}$$

Where in (1) and (2), $g(x) = x^8 + x^4 + x^3 + x^2 + 1$; this approach enables lossless image reconstruction without additional pixels.

### 2.2 RDH

#### 2.2.1 RDHEI

Scientists have proposed innovative methods for reversible data hiding in encrypted images (RDHEI), utilizing various techniques to enhance embedding capacity, optimize image quality, and improve data security.

Yin et al. (2022) proposed a RDHEI method based on pixel prediction and bit-plane compression. This method leverages pixel prediction to reduce data redundancy by predicting pixel values and calculating the error between the predicted and actual values. Regions with smaller errors are identified as embedding targets, optimizing the selection of embedding locations. Additionally, the method analyzes the distribution characteristics of bit planes in the image and compresses highly redundant bit planes to release more embedding space, thereby increasing embedding capacity without compromising image quality. Encryption ensures data privacy, while the separability of embedded information from the image enhances security and flexibility.

Yao et al. (2024) introduced a RDHEI technique combining global compression of zero-valued high bit planes with block rearrangement. By globally compressing zero bits in high bit planes, the method releases substantial embedding space. Simultaneously, block rearrangement optimizes the distribution of embedded data, achieving higher embedding efficiency and stronger data integrity. Furthermore, this method flexibly adjusts embedding strategies to significantly improve the utilization of image embedding space while ensuring the preservation of the original data during image recovery.

Yu et al. (2022) proposed a RDHEI method based on hierarchical embedding. This approach divides the embedding process into multiple levels, prioritizing regions with smaller prediction errors to minimize the impact on image visual quality. Additionally, the method introduces a hierarchical management mechanism, progressively embedding data and flexibly embedding different types of information in various regions, improving data management efficiency. The hierarchical embedding strategy ensures not only the reversibility of embedded data but also the independence of embedded information from the original content, providing reliable support for multi-role data management.

Yin et al. (2020) proposed a RDHEI method combining multi-MSB (most significant bits) prediction with Huffman coding. This technique predicts multiple MSBs for each pixel and analyzes the differences between the predicted and actual values to reduce redundancy. Huffman coding is then used to efficiently compress these differences, further optimizing embedding capacity. This combination of multi-bit prediction and compression improves embedding efficiency while ensuring precise recovery of data and reducing storage overhead, making it suitable for applications requiring high-precision data recovery.

Yu and Zhang (2023) proposed a method for reversible data hiding in encrypted images that combines secret sharing and hybrid coding. The approach uses bit-plane compression and difference expansion to optimize embedding efficiency and distributes embedded data across multiple shares for enhanced security. Secret sharing ensures that data recovery requires a minimum number of shares, protecting the integrity and confidentiality of the embedded information.

Despite the significant technological advances of these methods, they also share some common limitations. The embedding capacity of these methods depends on the distribution of image content. For images with complex textures or high-frequency content, the lack of redundancy may result in reduced embedding capacity.

#### 2.2.2 SS-based RDH

Scientists have proposed various reversible data hiding methods in encrypted images by integrating secret sharing techniques, focusing on improving embedding capacity, security, and data integrity.

Wu et al. (2018) proposed a reversible data hiding method in encrypted images using secret sharing. By dividing the image into multiple shares, the data can be embedded and later recovered using a minimum number of shares. This method ensures data security, effectively protecting the integrity and confidentiality of the embedded data, even if some shares are lost or compromised.

Qin et al. (2021) improved the secret-sharing-based reversible data hiding method by employing mathematical mechanisms in finite fields $GF(q)$ and $GF(2^8)$. Their approach enhances the efficiency of share distribution and utilization of image embedding space, increasing embedding capacity and recovery precision while maintaining data reversibility and privacy protection.

Chen et al. (2020) introduced a secret-sharing-based reversible data hiding method in encrypted images designed for multiple data hiders. This method allows multiple hiders to simultaneously embed different information into the encrypted image while preserving the reversibility of the original image. By enabling multi-role share design, the method enhances the flexibility and security of data management.

Hua et al. (2022) proposed a cipher-feedback-based secret sharing reversible data hiding method for encrypted images. By optimizing data distribution through a cipher-feedback mechanism, this method improves the robustness of embedded data against attacks and enhances the precision of image recovery. It is particularly suitable for applications requiring high security and embedding efficiency, such as healthcare and privacy protection.

Despite the significant advancements of these secret-sharing-based reversible data hiding methods, there are still some limitations. First, the utilization efficiency of embedding space requires further improvement, as some methods fail to fully exploit low-bit information or other potential redundancies in the image, resulting in underutilized embedding capacity. Second, from the perspective of embedding roles, these methods often assume single or limited roles for data embedding, lacking support for collaborative multi-role scenarios. For instance, in healthcare, patients, doctors, and hospitals may need to embed different types of data independently. However, current methods may face embedding conflicts between roles or fail to flexibly meet the security and priority requirements of different roles. Additionally, the data separation mechanism between multiple roles needs optimization to prevent role privilege leakage or misuse of data.

## 3 Proposed method

In modern healthcare systems, the secure storage and management of medical information remain important and complex issues. With the acceleration of digitalization, an increasing amount of medical information is stored and transmitted electronically, including patient identity information, medical history, diagnostic records, medical images, and more. The security of this data directly affects the protection of patient privacy and the improvement of healthcare quality. However, as different types of medical information have varying levels of importance, and different regions within medical images also vary in significance, it is essential to apply targeted, hierarchical protection to these data. For example, lesion areas (such as tumors or injuries) are especially critical in diagnosis and treatment, requiring higher levels of security than other regions.

*Patient's private information* is the most sensitive part of medical data, usually including identity information (such as name, ID number, address) and medical history (like past disease records and treatment history). Unauthorized disclosure of this information could pose significant privacy risks to patients, potentially impacting their social life and mental well-being. Therefore, private information should be under the strictest protection, and only accessible to the patient or authorized individuals. Based on this principle, private information should be held and managed by the patient themselves, rather than by medical institutions or third parties, to minimize the risk of data leaks.

*Medical reports* constitute another critical category of information, typically containing diagnostic results, treatment plans, and information about the attending physician. This information is crucial during the treatment process and directly impacts the formulation and execution of treatment strategies. Since medical reports are mainly used for communication and management within medical institutions, they can be managed by hospitals. As the primary holders of medical reports, hospitals can, under authorization, provide these reports to doctors, patients, or other authorized parties, ensuring the smooth progression of medical processes.

In practical healthcare applications, not all users require access to all types of medical information. When diagnosing, doctors typically only need to view specific areas of medical images, such as lesion areas, without needing access to the patient's identity and other private information. This role-based hierarchical access strategy not only allows doctors to focus on disease-related information but also effectively reduces the risk of exposing sensitive information, minimizing the possibility of privacy breaches. For example, during diagnosis, doctors only need to view the image data of the lesion area without knowing the patient's personal background or private information.

To achieve secure and flexible management of medical data access, we propose a reversible and separable multi-party healthcare framework to meet the distributed storage and secure access needs of medical information among multiple parties. The design of this framework is shown in Fig. 1. In this framework, the primary authorized parties include the *patient* and the *hospital*. The patient holds their private

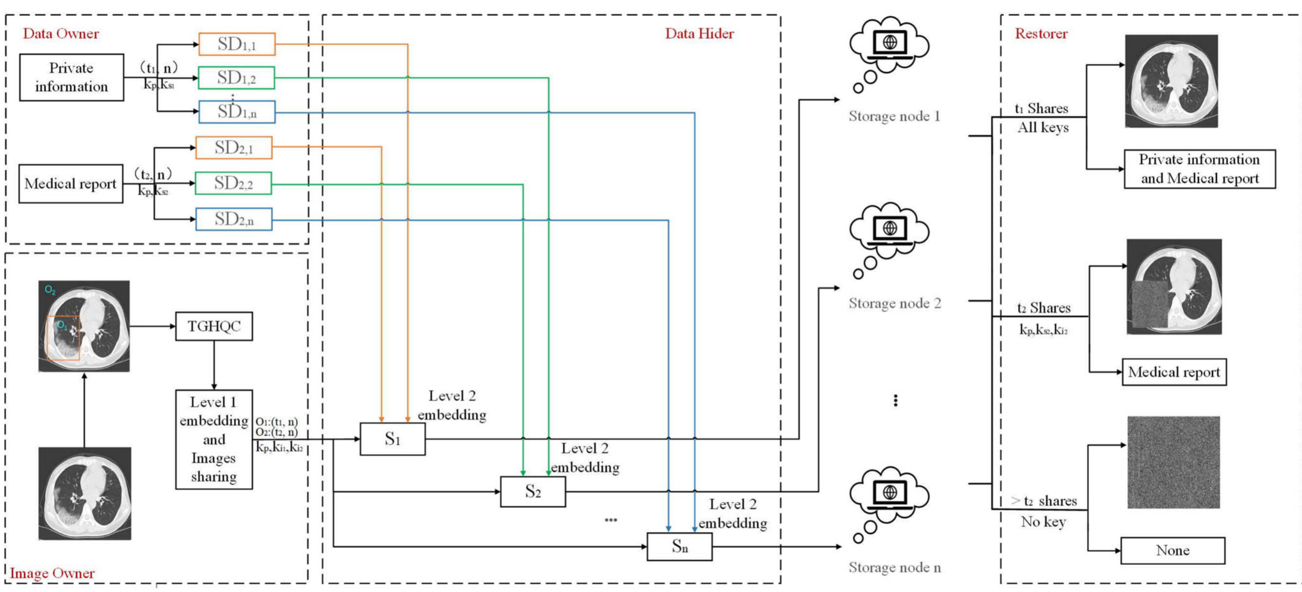


Fig. 1 Proposed reversible and separable multi-party healthcare framework

information and is responsible for protecting their identity information and medical history, while the hospital manages the medical report. This decentralized management approach not only protects patient privacy but also ensures effective management of diagnostic information by the medical institution.

The workflow of this framework is as follows: After the doctor completes the diagnosis, the hospital identifies and marks the lesion areas in the medical image. The original medical image is then preprocessed and shared as $n$ shares according to a threshold. During the sharing process, a series of medical image metadata, such as image generation device information, scanning parameters, and file format, are embedded in the redundant space of the polynomial coefficients in binary form. Subsequently, the system server embeds the shared information of private data and medical reports into different parts of the medical image according to the framework. Assuming there are $n$ cloud servers for data embedding, each cloud server obtains $k_p$, one share of the image, one share of the patient's private information, and one share of the medical report. Based on $k_p$, each server knows the specific location for data embedding, and then proceeds to embed the data sequentially. The lesion area image data will contain the shared data of the patient's private information, while the non-lesion area image data will contain the shared data of the medical report.

Based on this mechanism, the patient holds the decryption keys for the lesion image and private information (such as $k_{i1}$ and $k_{s1}$), while the hospital holds the decryption keys for the non-lesion image and medical report (such as $k_{i2}$ and $k_{s2}$). During the image recovery process, the patient has the highest level of authority and can decrypt all information. Without the patient's authorization, the hospital can only extract the non-lesion areas of the image and the diagnostic report, but cannot recover the complete image or the patient's personal information, effectively protecting the patient's privacy.

Additionally, due to the reversible and separable nature of this framework, it supports the option of recovering only the image data when only the image is needed, without extracting the patient's private information. For example, in research analysis, researchers may only need the image data for modeling purposes, without any patient identity information. In this scenario, the system ensures the anonymity of research data, further enhancing data security and compliance.

This framework, based on a multi-party reversible data hiding strategy, not only strengthens the security of the healthcare information system but also provides an efficient and flexible solution for storing and managing medical images. The design meets the varied data access needs of different roles, allowing patients, hospitals, doctors, and other stakeholders to access data according to their level of authority, thus achieving hierarchical management and protection of medical data. While safeguarding patient privacy, it optimizes the configuration of access permissions and offers robust support for the security and control of healthcare information systems.

## 3.1 Texture-guided hierarchical quantization coding

This section provides a detailed introduction to the proposed Texture-Guided Hierarchical Quantization Coding (TGHQC). This method aims to maximize embedding efficiency for medical images by analyzing their texture characteristics and adaptively adjusting data encoding and storage strategies, while ensuring the security and integrity of the data. The core concept of TGHQC lies in leveraging texture-based classification of image blocks and designing differentiated encoding approaches for different types of pixel blocks to optimize storage and embedding space utilization. TGHQC not only ensures the quality and integrity of medical image data during storage but also supports secure embedding and management of confidential information by integrating with the multi-party framework described in Section 3.1.

### 3.1.1 Texture-aware adaptive difference generation

Assume that $I$ is an 8-bit original medical image with a size of $M \times N$, which is divided into several non-overlapping $w \times l$ blocks. all blocks within image have three types of difference generation methods: prediction error (PE), median difference (MND), and maximum difference (MMD).

To improve the performance and effectiveness of the algorithm, we automatically select the difference generation method by evaluating the texture characteristics of pixel blocks. A unified formula is defined based on the statistical properties of image textures, which can accurately detect edge pixels in a given image using the Canny operator (Canny 1986). For a given block, if it contains very sparse edge pixels, it can be considered a smooth block, labeled as PE. On the other hand, if it contains many edge pixels, it means that this block has a lot of high-frequency energy and can be considered a texture block, labeled as MND. If the edge pixel density is moderate, indicating that the area contains a considerable amount of edge information, it is labeled as MMD. Thus, blocks can be classified according to the density of edge pixels.

As we know, the Canny operator is a very famous and powerful edge detector that can accurately detect edge pixels in a given image. The images in the middle column of Fig. 2 show the edge maps detected by the Canny detector. For a given block, if it contains very sparse edge pixels, it

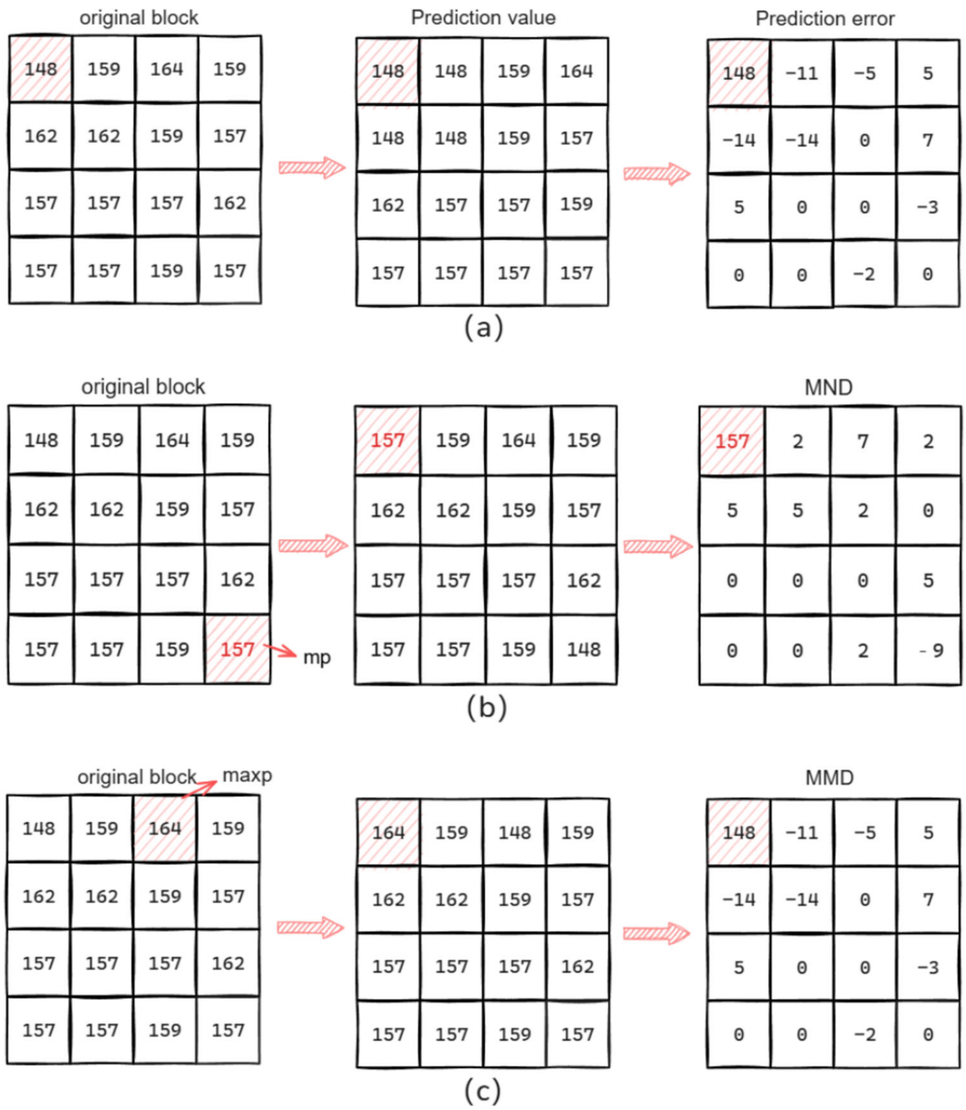


**Fig. 2** Difference generation: (a) PE generation, (b) MND generation, (c) MMD generation

can be considered a smooth block. On the other hand, if it contains many edge pixels, it means that this block has a lot of high-frequency energy and can be considered a texture block. Thus, blocks can be classified based on the density of edge pixels.

Based on the above analysis, a parameter named edge pixel density $\rho_{edgel}$ is defined as

$$\rho_{edgel} = \sum_{edgel} /(w \times l) \tag{3}$$

where $\sum_{edgel}$ is the number of edge pixels in a given block within the edge map generated by the Canny operator, $l$ is the length of block. The block type is determined by (4). We set $\alpha = 0.1$ and $\beta = 0.2$. Figure 2 shows the examples of the block classification method:

$$\text{Block type} = \begin{cases} \text{PE}, & \rho_{edgel} \leq \alpha \\ \text{MMD}, & \alpha < \rho_{edgel} \leq \beta \\ \text{MND}, & \rho_{edgel} > \beta. \end{cases} \tag{4}$$

After classifying the pixel blocks, the corresponding differences need to be generated. The generation methods for various differences are as follows.

*PE* Let $k_{i,j}$ represent the pixel value at position $(i, j)$, where $k_{i,j} \in [0, 255]$, $i \in [1, w]$, $j \in [1, l]$. Due to the high correlation between adjacent pixels in the original image, these pixels can be predicted using a predictor based on their contextual pixels, such as the Median Edge Detection (MED) predictor. Let $p_{i,j}$ be the predicted value of $k_{i,j}$. For the pixel located in the top-left corner of the original image, its predicted value is its actual pixel value.

$$p_{1,1} = k_{1,1} \tag{5}$$

For other pixels that are not in the top-left corner, they are predicted using the extended MED as follows:

$$p_{i,j} = \begin{cases} b, & \text{if } i = 1 \text{ and } 2 \le j \le l \\ a, & \text{if } 2 \le i \le w \text{ and } j = 1 \\ \max(a,b), & \text{if } 2 \le i \le w, 2 \le j \le l, \\ & c \le \min(a,b) \\ \min(a,b), & \text{if } 2 \le i \le w, 2 \le j \le l, \\ & c \ge \max(a,b) \\ a+b-c, & \text{otherwise} \end{cases} \tag{6}$$

where $a = k_{i-1,j}$, $b = k_{i,j-1}$, $c = k_{i-1,j-1}$. Then, the prediction error (PE) $e_{i,j}$ is calculated as:

$$e_{i,j} = k_{i,j} - p_{i,j} \tag{7}$$

where $e_{i,j} \in [-255, 255]$. Figure 2 (a) shows the PE generation process of the original pixel block.

*MND* The sixteen pixel values within each block are first filtered, and a labeled pixel $mp$ is selected so that it satisfies the condition in (8) and exchanges position with the first pixel in the block. Figure 2 (b) shows the MND generation process of the original pixel block.

$$mp = min(max \mid mp - x_n \mid), n = 1, 2, ..., w \times l \tag{8}$$

where $x_n$ represents the $n$-th pixel value during the row-wise traversal of the pixel block.

*MMD* Similar to the generation of the labeled pixel $mp$, the $w \times l$ pixels within each block are first filtered, and according to (9), the maximum value among the $w \times l$ pixels is selected as the reference pixel $maxp$ and is swapped with the first pixel. Figure 2 (c) shows the MMD generation process of the original pixel block.

$$maxp = max(x_n), n = 1, 2, ..., w \times l \tag{9}$$

where $x_n$ represents the $n$-th pixel value during the row-wise traversal of the pixel block.

### 3.1.2 Hierarchical quantization coding

We designed Hierarchical quantization coding, combined with the texture-aware adaptive difference generation described in the previous section, to achieve maximum data embedding. After performing texture-aware adaptive difference generation, the generated corresponding differences are converted into binary form, represented as the set of binary lengths $L_{d_i}$. Then, based on these lengths, the elements in $L_{d_i}$ are divided into $k$ levels, with the corresponding difference length being $L_k$. The maximum value of $k$ is clearly 7. Assuming a block has $k$ levels; if $k = 1$, the block belongs to the original difference set $O_{d_i}$. If $k = 2$, and the difference lengths of the first and second levels are 1 and 2 respectively, the block still belongs to $O_{d_i}$. All other cases are classified as partially shortened differences $PS_{d_i}$. For $O_{d_i}$ blocks, no further processing is performed. However, for $PS_{d_i}$ blocks, only the differences at the $k$th level are processed, assuming that all differences $\{d_m\}$ in the $k$th level have been obtained. The processing method for $\{d_m\}$ is shown in the following equation:

$$d'_m = \begin{cases} \frac{d_m}{2^n}, d_m \ mod\ 2 = 0 \\ \frac{d_m - 1}{2}, d_m \ mod\ 2 \ne 0 \end{cases} \tag{10}$$

Where $d_m \in \{d_m\}$;

For $d_m$, the operation described in (10) needs to be repeated $n$ times, where $n = L_k - L_{k-1}$. For $PS_{d_i}$ blocks, if $n = 1$ and the number of $d_m$ elements in the $k$th layer is too large, the encoded length may exceed the original length before encoding, which would defeat the purpose of hierarchical quantization coding. Based on experimental verification, we have established specific limits $q_n$ for the number of $d_m$ elements in the $k$th layer corresponding to different $n$ values. If the number exceeds $q_n$, hierarchical quantization coding should not be used, and Huffman coding should be attempted instead. The relevant specifications are shown in Table 1.

While applying hierarchical quantization coding, we also encode the block differences using Huffman coding and compare the resulting length with that of hierarchical quantization coding, selecting the method with the shortest encoding length and making the corresponding records. We generate PE, MND, MMD of the original image for individual difference generation methods respectively, coding of the block level differences are generated according to the huffman coding of the differences of image.

While performing hierarchical quantization coding, we also generate a Huffman table for the difference values of the original blocks across the entire image according to

**Table 1** hierarchical quantization coding constraints

| n | $q_n$ |
|---|---|
| 1 | 1 |
| 2 | 3 |
| 3 | 5 |
| 4 | 6 |
| 5 | 7 |

different block difference generation methods. Then, the corresponding blocks are encoded. We compare the lengths of the hierarchical quantization coding encoding and the Huffman encoding, select the shortest method, and record it.

#### 3.1.3 Block-level auxiliary information generation

The auxiliary information within all types of blocks is divided into two categories: Encoded Marking data($EM$) and Difference Sign recording data($DS$). The encoded marking information is roughly divided into the following categories:

$$Au = L_s \mid\mid MS \mid\mid L_p \mid\mid B_M \mid\mid n \mid\mid num \mid\mid L_m \mid\mid F \mid\mid Rp \tag{11}$$

1. Determine the embedding position $L_s$: This is used to find the starting position for embedding secret data in the shared $MP$ block, occupying 4 bits. This part is not involved in the encoding process.

2.Mode selection information$MS$: After selecting the optimal difference generation method, we need to use 2 bits to mark the chosen method: PE is '00', MND is '01', and MMD is '10'. After generating the differences, we perform TGHQC encoding. If the length of the differences encoded by Huffman is greater than that of TGHQC encoding, TGHQC encoding is used, and the auxiliary information is marked as '0'; otherwise, it is marked as '1'. Therefore, the total auxiliary information for mode selection is 3 bits.

3. Mark pixel position $L_p$: For the PE method, it is not necessary to mark pixel position information, so if the first two bits of the extracted scheme choice information are '00', $L_p$ can be ignored. For the other two block types, we determine the position of the marked pixel $mp$ or maxp within the block, ranging from '0000' to '1111', occupying 4 bits.

4. Maximum difference bit count $B_M$: The binary bit length of the maximum difference within the block is calculated and represented using 3 bits.

5. $n$, $num$, $L_m$, and $F$: These four parts represent the information about the original differences and are introduced together. $n$ indicates the number of cycles required for $d_m$, and since the maximum value of $n$ is 6, it is fixed at 3 bits. $num$ represents the number of elements in $\{d_m\}$, and the maximum value of $q_n$ is 7, so it is also fixed at 3 bits. $L_m$ represents the specific positions of $d_m$, where each $d_m$ is represented by 4 bits, so $L_m$ occupies $num \times 4$ bits. $F$ indicates whether each $d_m$ has a minus 1 operation in each cycle; if it does, $F$ is 1, otherwise it is 0. The length of $F$ is determined by $n$ and $num$, with a total of $n \times num$ bits. Therefore, the length of this auxiliary information depends on the situation, and its length can be calculated as $n + num + n \times num$ bits.

6. Reference pixel $R_p$: Although blocks are divided into different types, the length of the reference pixel is uniformly 8 bits for all types.

For a PE block, the total length of the encoded marker information is calculated as $18 + n + num + num \times 4 + n \times num$bits. In contrast, for the other two block types, this length is $22 + n + num + num \times 4 + n \times num$ bits. The encoded marking information is further categorized into several distinct components, such as $Ls$, $MS$, $Lp$, $BM$, $n$, $num$, $Lm$, $F$, and $Rp$.

Next is the difference symbol recording information. For PE and MND blocks, the differences can be positive or negative, so it is necessary to record the sign of the interpolated values for pixels other than the reference pixel, requiring 15 bits. In the MMD block, all values are smaller than the reference pixel, so no sign needs to be recorded. If the value is greater than $mp$, it is recorded as '1', otherwise, it is recorded as '0', taking up a total of 15 bits. The difference information and the Huffman table are then stored into the image as auxiliary information for the entire image, block by block. The same processing method is applied to the other regions. After processing all regions, the image proceeds to the sharing step.

### 3.2 Level 1 embedding and polynomial-based sharing

The processed image is first segmented into two regions using the method in Yang et al. (2020), based on the parts defined by the hospital, and the location coordinates of each region are recorded as $lo_1, lo_2$. These coordinates $lo_1, lo_2$ are used as part of the public key for recovering the original image. The corresponding image regions are labeled as $I_1, I_2$, and assigned ranks $G_1, G_2$, with the ranks decreasing sequentially.

This subsection will explain the sharing algorithm process in detail, as shown in Algorithm 1. After the image owner processes the image using the encoding method proposed in this paper, the image encryption keys $ki_1, ki_2$, along with thresholds $(t_1, n_1)$ and $(t_2, n_2)$, are used as conditions on the image regions $I_1, I_2$ to generate $n_1, n_2$ shadow images $s_1, s_2, ..., s_n$ through secret image sharing. Similarly, the private information and medical reports owned by the patient and doctor, represented as $SD_1, SD_2$, are shared using the data encryption keys $ks_1, ks_2$ and thresholds $(t_1, n_1), (t_2, n_2)$, where $t_1 \geq t_2$.

Since the encoding method in this approach is designed for pixel blocks, when image sharing is applied to each image block, we first convert every 8 bits in $Au$ into decimal numbers, resulting in $\sigma_1, \sigma_2$,.... For a polynomial with (k, n) threshold, $n$ polynomials of degree $k - 1$ are required, resulting in a total of $k \times n$ polynomial coefficients. However, sometimes the compressed original image information is insufficient to fill $k \times n \times 8$ binary bits. Therefore, for

**Algorithm 1** Proposed sharing algorithm

**Input:** Secret image $I$, secret data $SD$, thresholds $t_1, t_2, (t_1 \geq t_2), n_1, n_2$;
**Output:** shares $S_i, i = 1, 2, ..., n$;
$I$ is partitioned into different regions $I_1$, $I_2$, and the location coordinates of the regions are recorded as $lo_1$, $lo_2$. $SD$ is partitioned into $SD_1$, $SD_2$, and accordingly, the ranks $G_1$, $G_2$ are assigned respectively, which decrease sequentially.
**for** $SD_i, i = 1, 2$ **do**
  The secret sharing algorithm in (1) with threshold $(t_i, n_i)$ generates $SD_{(i,1)}, SD_{(i,2)}, ..., SD_{(i,n_i)}$
**end for**
**for** Image $I_i, i = 1, 2$ **do**
  Using the threshold $(t_i, n_i)$ as the condition, share to generate shadow images $s_i, i = 1, 2, ..., n$
**end for**

the redundant coefficients, we substitute them with medical image metadata,which is the Level 1 embedding. Once these steps are completed, we can share the block according to (1) and the image sharing key $k_i$.

### 3.3 Level 2 embedding

After the image owner completes the Level 1 embedding and generates the image shares for uploading to the cloud server, the data embedder performs the Level 2 embedding. Data embedder is responsible for embedding the patient's private information and medical reports into the corresponding locations within the shared images. This dual embedding method optimizes the utilization of space within each image share.

### 3.4 Data extraction and image recovery

A minimum of $t_k$ authorizers with level $G_k$ are required to receive and manage the shared data in order to extract the data. Then, the extracted $t_k$ data are recovered into secret data $SD_k$ using the Lagrange interpolation algorithm.

To extract the embedded information and recover the original secret image, the recipient first extracts the data structure (DS) and the Huffman coding table based on the key. Then, each received share is divided into 4×4 blocks, and regions $O_1$, $O_2$,..., $O_m$ are distinguished using the public key $k_p$. The recipient extracts the data and recovers the image according to the level and type of keys they possess.

Using the image decryption key $k_i$ and the public key $k_p$, the original image can be directly reconstructed without extracting the secret data. Using the data hiding key $k_s$ and the public key $k_p$, the secret data can be directly extracted.

Case 1: $k_i$ and $k_p$

Extract the first pixel value from each $w \times l$ shadow pixel block and substitute it into the Lagrange interpolation polynomial in (2). Simplify the polynomial and extract the constant coefficient $EM_1'$. Convert this coefficient to binary and read the first four bits to determine the end position $L_s$ of the original image information. Next, based on the position information and $k_i$, sequentially extract the image information. Since the length of the auxiliary information depends on the block type, the block type must be determined while extracting the information.

After reading the first four bits of the auxiliary information, continue by reading the next three bits $SC$, which indicate the block type and whether TGHQC encoding has been applied. If no encoding has been applied, directly look up the table to read the Huffman code and restore the original difference information. If it is a PE block, there is no need to obtain the original position of the reference pixel; otherwise, read the next four bits to obtain $L_p$. Then, obtain the binary bit length of the differences $B_M$, $n$, and $num$. Based on the values of $B_M$ and $n$, the length of the difference information can be inferred as $(B_M - n) \times 15$ bits.

Next, extract $num$ to determine the number of modified differences, then extract the subsequent $num \times 4$ bits to determine the positions of these differences. The following $n \times num$ bits correspond to the information about the difference changes. Finally, obtain the eight-bit reference pixel $R_p$ and the $(B_M - n) \times 15$ bits of difference information. Then, recover the original differences according to Section 3.4, thereby restoring the original image.

$$d_m = d_m' \times 2 + F_m(i), \quad m = 1, 2, \ldots, num, \quad i = 1, 2, \ldots, n$$

Case 2: $k_s$ and $k_p$

Using $k_p$, reconstruct the $L_s$ pixel, removing its auxiliary information while preserving the secret information. This secret information, which is reconstructed by $k_s$, is then combined with the previously extracted information to form the complete secret data $SD$.

Case 3: $k_i$, $k_s$, and $k_p$

The secret data $SD$ can be extracted first based on $k_s$, or the original image can be recovered first based on $k_i$. This scheme is both reversible and separable.

### 3.5 Numerical example

To better understand the algorithm, we demonstrate the proposed method using the sample block shown in Fig. 2.

First, according to (3) and (4), we can determine that the sample 4 × 4 block belongs to the MND category. The next step is to perform hierarchical quantization coding. Taking (4, 3) sharing as an example, according to Table 1, we conclude that this block meets the conditions for hierarchical quantization coding. As shown in Fig. 3, according to (8), the marked pixel $mp$ for this block is 157, and the difference array $d$ between the marked pixel and the other pixels is calculated as [5 0 0 2 7 0 0 7 2 0 2 2 2 0 5 9]. After performing the hierarchical quantization coding operation, the differences become [5 0 0 2 7 0 0 7 2 0 2 2 2 0 5 4], with a maximum

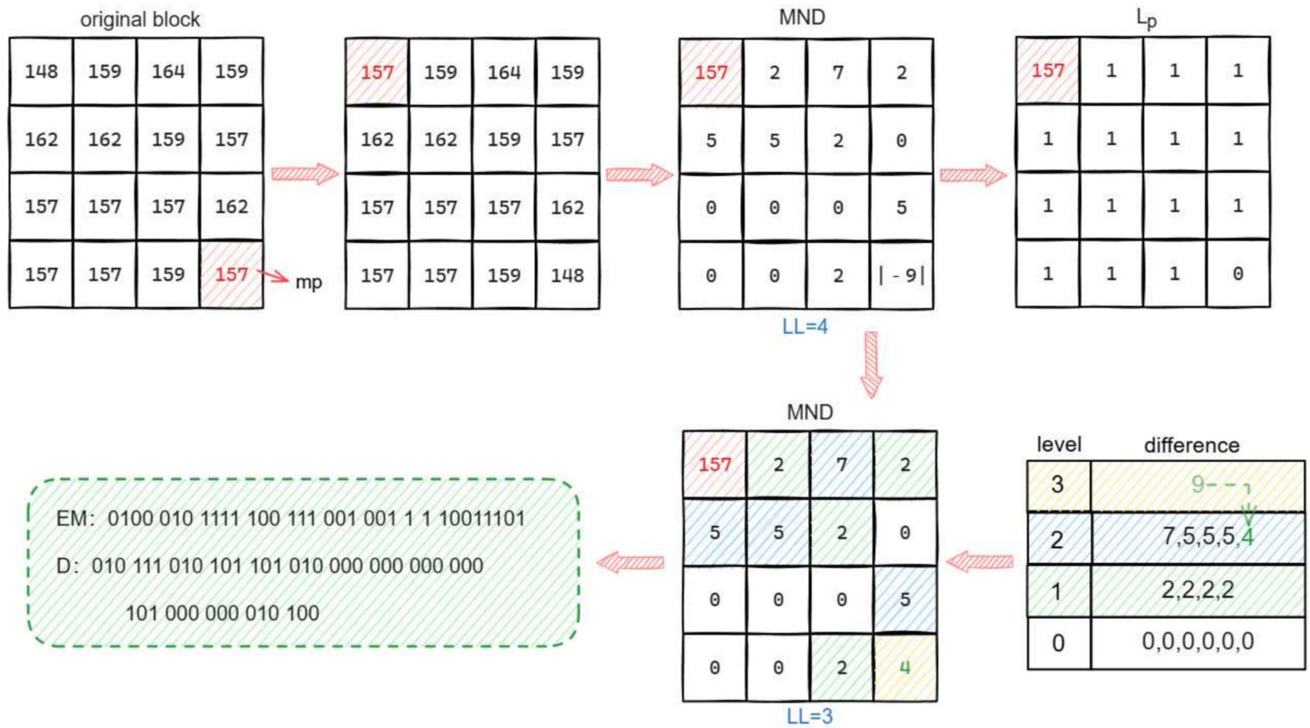


**Fig. 3** Texture-guided hierarchical quantization coding

value of $(7)_{10} = (111)_2$. Therefore, the corresponding number of binary bits $B_M$ is $(3)_{10} = (011)_2$. The difference sign information is recorded as $(100110011011010)_2$, where values greater than 157 are labeled as "1", and those less than or equal to 157 are labeled as "0". Since this is a marked block, $M_s$ is "1". The position of the marked pixel $mp$ in the block $L_p$ is "1111", and the position $L_m$ in the difference block is also "1111", with a corresponding value $n$ equal to 1. Therefore, the change feature $F$ is only 1 bit and is set to 1. Additionally, by converting each decimal number in the array $d_i', i = 1, 2, ..., 15$ into a 3-bit binary form, constrained by the limit of $B_M$, the auxiliary information set $D$ for this block is generated. $L_s$ is calculated as $\lceil \frac{78}{24} \rceil = (4)_{10} = (0100)_2$. The specific values are shown in the figure and will be used in the subsequent sharing phase. Since this block is classified as an MND block, the first two bits of $MS$ are 01. The Huffman encoding length for this block is 81 bits, so hierarchical quantization coding is still selected, and $MS$ is 010.

As shown in Fig. 4, the process of generating shared data for the sample block is demonstrated. Assuming the sharing threshold is (3, 4), after generating the auxiliary information and difference information in Fig. 3, we first arrange the auxiliary information, followed by the difference information. These are divided into several 8-bit binary sequences and converted into decimal numbers less than or equal to 255. These numbers are used as polynomial coefficients, which are then divided into four parts for sharing based on $k_i$. Before embedding, we pre-calculate the space occupied by all auxiliary information after sharing, which is $L_s = 4$. Therefore, this information occupies only four pixels in each share, with each pixel sharing 24 bits of information, meaning that four pixels require 96 bits. However, in Section 3.1.3, we calculated the total length of $Au$ and $d_i'$ to be 78 bits, leaving 18 bits of blank space. To fill this blank space and further utilize redundant space, we plan to Level 1 embedding. The remaining space cannot be fully utilized in Fig. 4, because

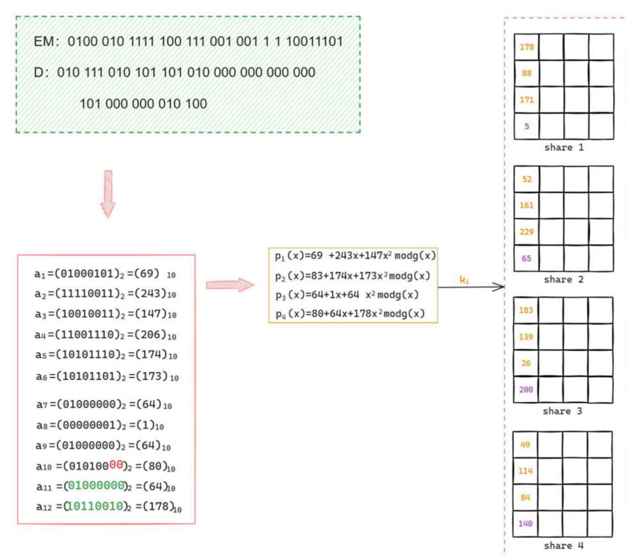


**Fig. 4** Blocks (3, 4) Sharing of MND

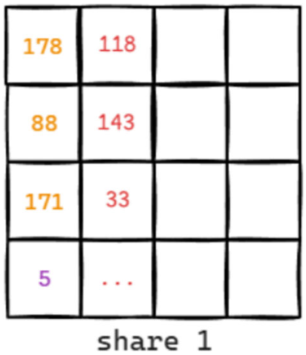


**Fig. 5** Embedding of share 1

replacing the first two bits of the redundant space would prevent medical image metadata from being restored. Therefore, only the last 16 bits can be replaced with useful information, which contain the medical image metadata.

Figure 5 illustrates the Level 2 embedding process for Data Hider 1 in Share 1. After receiving Share 1, Hider 1 first uses $k_p$ to determine the sharing threshold of the block, thereby identifying the data embedding position. The secret data is then sequentially embedded into the blank pixel blocks according to the sharing order.

Figure 6 provides a detailed example of extracting data from the shares and recovering the original pixel values. As shown in this figure, the recipient receives any three shadow blocks: Share 1, Share 2, and Share 3. By using only the image decryption key and the public key, the first pixel value can be extracted from each shadow image block, determining that the image information is located in the fourth pixel position. After determining that the recorded information in the entire block is 78 bits long, the corresponding difference and auxiliary information are extracted. For pixels encoded using hierarchical quantization coding, the receiver sequentially restores the original differences based on Section 3.4, thereby recovering the original pixel blocks. If only the data extraction key and the public key are available, the recipient can determine the starting position of the embedded secret data based on the public key information, allowing for the complete extraction of the secret information.

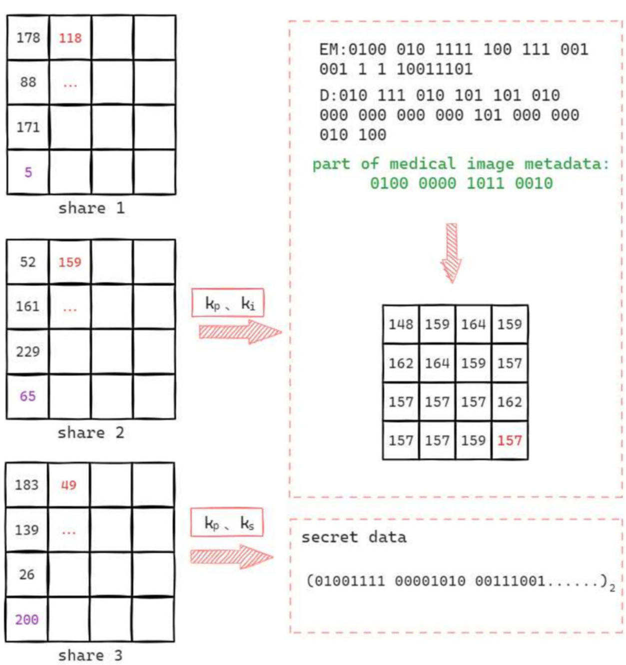


**Fig. 6** Data extraction and block recovery

## 4 Experimental results and analysis

In this section, we present the experimental results of the algorithms. All medical test images shown in Fig. 7 are selected from COVID-CT (Yang et al. 2020). Additionally, we also selected seven classic standard images for comparison experiments, and all images are of size 512 × 512.

### 4.1 Security analysis

*Visual Security*: Fig. 8 provides an example of the image sharing and extraction process of Fig. 7 (d) using the RDHSI algorithm, with thresholds of (2, 4) and (3, 4), w = 2, l = 4. In this figure, $SD_1$ and $SD_2$ represent private information and medical reports, respectively. The information is embedded into shares $S_1$ through $S_4$. The recipient, the patient, can use the key to recover both the lesion and personal information;

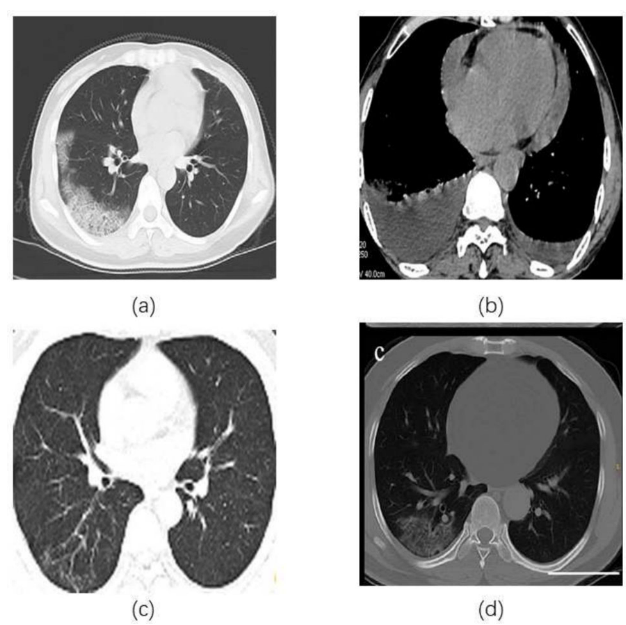


**Fig. 7** Text images (Yang et al. 2020)

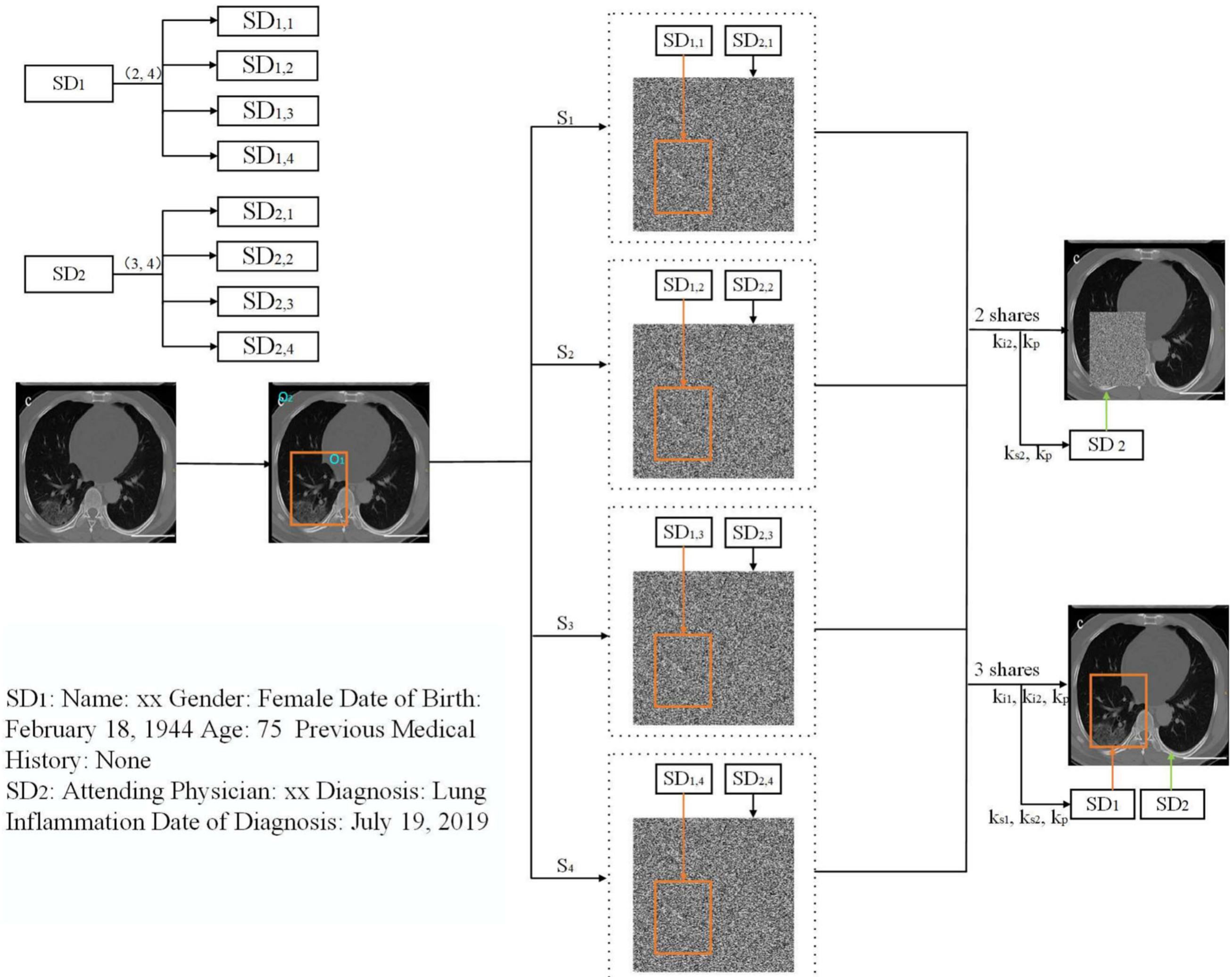


**Fig. 8** Example of recovery and extraction in medical images shown in Fig. 7 (d)

however, without the patient's authorization, the hospital can only recover the non-lesion area and the medical report.

From this, it can be observed that all shared images appear as noise images and do not contain any discernible information from the original image, making the embedded data visually imperceptible. These shared images do not reveal any details about the original image or the secret data. For an attacker, the content of the original image and the embedded data remains undetectable.

Therefore, it can be concluded that the proposed scheme is lossless. Additionally, the size of the generated shared images is equal to the size of the original image. From this, we can conclude that the proposed scheme is both secure and reliable.

*Image Histogram*: Fig. 9 shows the histogram of the original image in Fig. 8, along with the histograms of the four shares and recover image. It is clear from the figure that there is a significant difference between the histogram of the original image (a) and the histograms of the shared images (b), (c), (d) and (e). The histogram of the original image exhibits a more concentrated distribution, whereas the histograms of the shared images display a more uniform distribution, resembling noise. This indicates that during the sharing process, the original information of the image is effectively concealed, and the generated shared images do not contain any discernible features of the original image, thereby ensuring the security and confidentiality of the image sharing process.

*PSNR and SSIM*: To verify the visual security and encryption effectiveness of the proposed method during the information hiding process, we calculated the PSNR and SSIM values between the encrypted images (shared images) and the original images. PSNR evaluates the pixel-level similarity, while SSIM measures the structural similarity. Ideally, encrypted images should exhibit low PSNR and SSIM values close to 0 compared to the original images, ensuring that the

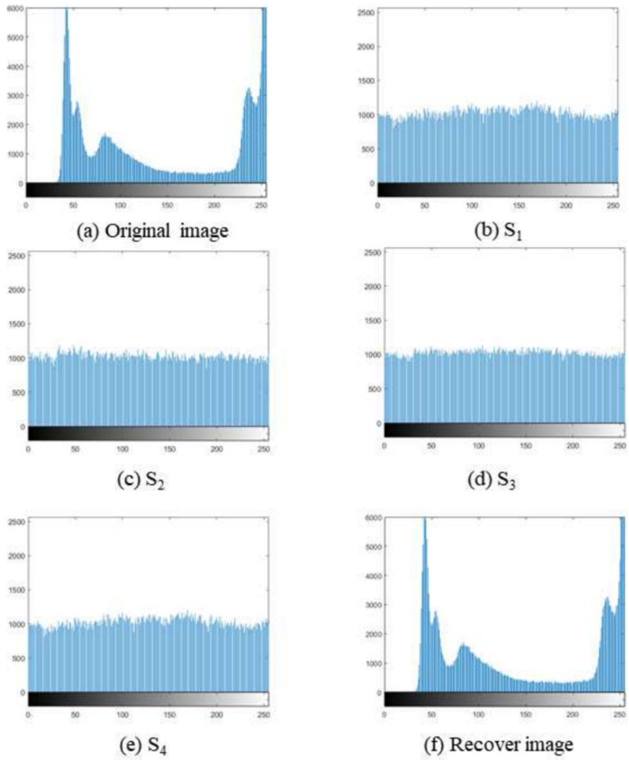


**Fig. 9** Histogram of original image, $S_1$-$S_4$ and recover image in Fig. 8

encrypted images are visually imperceptible and incapable of leaking original information.

In the experiment, medical images in Fig. 7 were selected as test images, and they were shared using a (3, 4) threshold scheme. The PSNR and SSIM of the reconstructed images and the four shares relative to the original images were tested. The results are shown in Table 2.

From the table, it can be observed that the PSNR values of all encrypted images are low (around 9 10 dB), indicating minimal pixel-level similarity between the encrypted images and the original images. Additionally, the SSIM values of the encrypted images are close to 0, showing that the encrypted images lack visual and structural similarity to the original images. This demonstrates that the encryption process effectively conceals the information of the original images, ensuring the visual security of the encrypted images.

The experimental results demonstrate that the proposed method can generate encrypted images with pseudo-random distributions, making them visually resemble noise. As a result, it is impossible to infer any information about the original images from the encrypted images. This further validates the security and reliability of the proposed method during the information hiding process.

## 4.2 Data expansion

According to the discussion in the literature (Wu et al. 2018), data expansion means that the shared encrypted image is larger than the original image. The data expansion rate of the entire scheme can be reflected by the ratio of the number of pixels in the shared encrypted image to the number of pixels in the original image.

$$\text{Expansion rate} = \frac{\text{number of shared image pixels}}{\text{number of original image pixels}} \tag{12}$$

Therefore, the expansion rate is used in the experiment to reflect the data expansion of the scheme, and the data expansion rate is expected to be as small as possible. The data expansion of this paper's scheme is compared with several different schemes proposed by Wu et al. (2018), Qin et al. (2021), Chen et al. (2019), Chen et al. (2020), and Hua et al. (2022), as shown in Table 3. In the schemes (Wu et al. 2018; Qin et al. 2021; Chen et al. 2019), and (Chen et al. 2020), the shared shares generated are the same size as the original image, so the expansion rate is 1. The size of the shared share generated by the scheme in Hua et al. (2022) is $1/(t$ - $1)$ of the original image, where $t$ is the recovery threshold, so its expansion rate is $1/(t$ - $1)$, showing better data expansion performance. In the scheme of this paper, the shared shares generated by the proposed sharing and embedding algorithm are the same size as the original image, the shared shares in multi-party recovery are the same size as the divided region, and the maximum size of the shared share is the size of the original image. Therefore, the maximum expansion rate of the proposed scheme in this paper is 1, demonstrating good data expansion performance.

**Table 2** PSNR and SSIM of original, shares, and recovered images of Fig. 7

| Images | $S_1$ | | $S_2$ | | $S_3$ | | $S_4$ | | Recovered Image |
|---|---|---|---|---|---|---|---|---|---|
| | PSNR | SSIM | PSNR | SSIM | PSNR | SSIM | PSNR | SSIM | |
| Fig. 7(a) | 9.87 | 0.011 | 10.12 | 0.014 | 9.95 | 0.013 | 10.03 | 0.010 | PSNR=$\infty$, SSIM=1.000 |
| Fig. 7(b) | 9.65 | 0.010 | 9.89 | 0.012 | 10.02 | 0.011 | 9.84 | 0.009 | PSNR=$\infty$, SSIM=1.000 |
| Fig. 7(c) | 9.78 | 0.012 | 10.05 | 0.013 | 9.92 | 0.010 | 9.99 | 0.008 | PSNR=$\infty$, SSIM=1.000 |
| Fig. 7(d) | 9.56 | 0.009 | 9.81 | 0.011 | 9.74 | 0.010 | 10.08 | 0.009 | PSNR=$\infty$, SSIM=1.000 |

**Table 3** Data expansion of different programmes

| Shames | Expansion rate |
|---|---|
| Wu et al. (2018) | 1 |
| Qin et al. (2021) | 1 |
| Chen et al. (2019) | 1 |
| Chen et al. (2020) | 1 |
| Hua et al. (2022) | $1/(t-1)$ |
| Proposed method | $\leq 1$ |

### 4.3 Payload capacity

In the experiments, we tested the embedding rate (ER) of the encoding scheme proposed in this paper under different sharing thresholds. All schemes measured the effective ER, which is the ER of effective data after removing the embedded auxiliary information, as shown in (12). We randomly selected several medical images, as shown in Fig. 7, and tested the ER of these images under several different sharing thresholds, with the results shown in Table 4. We set different $w$ and $l$ with different $t$. As the value of $t$ in the sharing threshold increases, more auxiliary information data can be accommodated in the polynomial, providing more space for embedding effective secret data in the image. Therefore, for the same image, the higher the $t$ value in the sharing threshold, the higher the ER.

To further enhance the persuasiveness of the experiments, we selected several commonly used classic images and compared them with various advanced RDHEI and RDHSI methods (Yin et al. 2022; Yao et al. 2024; Yu et al. 2022; Yin et al. 2020; Yu and Zhang 2023), we set $w = 4$, $l = 4$; As shown in Fig. 10, in the vast majority of classic images, our embedding rate is significantly higher than almost all other schemes, with the only exception being the airplane image, where it is slightly lower than (Yu and Zhang 2023). This demonstrates that the proposed method achieves a higher embedding capacity while maintaining image quality.

We also compared the TGHQC proposed in this paper with Yin et al. (2022); Yao et al. (2024); Yu et al. (2022); Yin et al. (2020); Yu and Zhang (2023) on two classic datasets, BOSScase and BOWS-2. The comparison results are shown in Fig. 11. The proposed TGHQC achieved significantly higher average embedding efficiency on BOSSBase and BOWS-2 than other schemes. It can be seen that the proposed TGHQC achieves a higher embedding rate compared to the other methods, demonstrating its superiority. Additionally, we also evaluated the embedding rate of TGHQC in the medical image dataset (Yang et al. 2020). Experimental results show that the proposed encoding method achieves an average embedding rate of 4.5433 bpp in the medical image dataset, indicating a relatively high embedding capacity.

**Table 4** Embedding rate of different test images of Fig. 7 in different shared thresholds

| image | (2, 4) Shared | (3, 4) Shared | (4, 4) Shared |
|---|---|---|---|
| | $w$=2, $l$=4 | $w$=2, $l$=4 | $w$=2, $l$=4 |
| | ER(bpp) | ER(bpp) | ER(bpp) |
| (a) | 3.65 | 4.59 | 5.56 |
| (b) | 3.91 | 4.85 | 5.87 |
| (c) | 4.56 | 5.40 | 6.37 |
| (d) | 3.72 | 4.64 | 5.59 |
| Average | **3.96** | **4.87** | **5.85** |

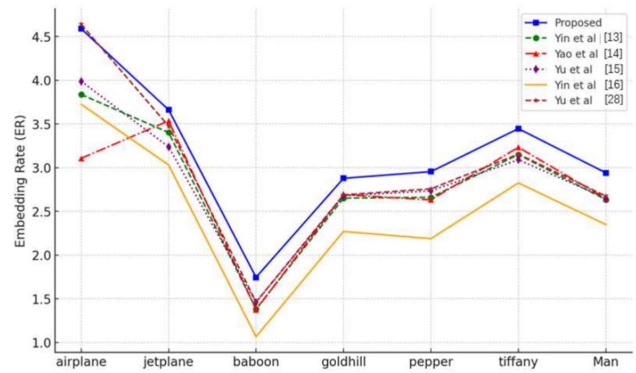


**Fig. 10** Comparison of ER with several advanced encoding methods in classic images

### 4.4 Performance comparison

This subsection summarizes the characteristics of several different related schemes and the results are shown in Table 5. The schemes in the literature (Wu et al. 2018; Qin et al. 2021; Chen et al. 2020; Yu and Zhang 2023) and (Hua et al. 2022) all address the case where the storage party needs to classify and label the image after the secret image is shared, so the data embedding is done in the cloud service provider or data hiding party. The algorithm proposed in this paper, on the other hand, addresses the situation where the system needs

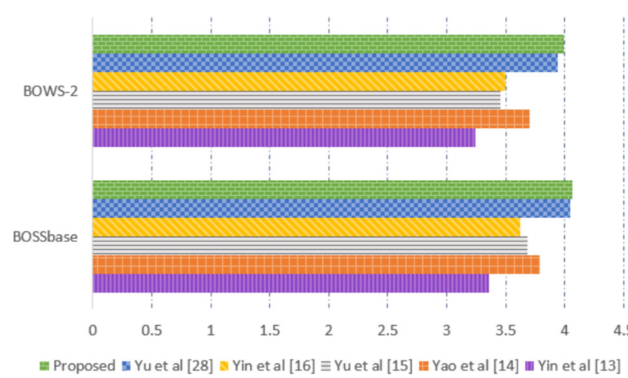


**Fig. 11** Comparison of ER with several advanced encoding methods in BOSScase and BOWS-2

**Table 5** Comparison of the performance of different shames

| Shame | Data hider | Secret data shared | Complexity of the algorithm | Level of access | Separable |
|---|---|---|---|---|---|
| Wu et al. (2018) | Single data hider | NO | $\frac{(t+2)(t-1)MN}{2}$ | NO | NO |
| Qin et al. (2021) | Single data hider | NO | $\frac{(t+2)(t-1)MN}{2}$ | NO | NO |
| Chen et al. (2020) | Multiple data hider | NO | $\frac{(t+2)(t-1)MN}{2}$ | NO | YES |
| Hua et al. (2022) | Multiple data hider | NO | $\frac{(t+2)MN}{2}$ | NO | NO |
| Yu and Zhang (2023) | Multiple data hider | NO | $\frac{(t+2)(t-1)MN}{2}$ | NO | NO |
| Proposed method | **Multiple data hider** | **YES** | $<\frac{(t+2)(t-1)MN}{2}$ | **YES** | **YES** |

to encrypt both the secret image and the additional data, so it chooses to carry out the sharing and data embedding operations in the content owner, which saves the transmission bandwidth and avoids the additional database access. The properties that each scheme relies on when embedding data are listed in the table.

The proposed scheme in this paper utilizes texture information of pixel blocks to generate differences, which maximizes space savings for data embedding. Additionally, the algorithms in Wu et al. (2018); Qin et al. (2021); Chen et al. (2020) and Yu and Zhang (2023) require sequential processing of each pixel value in an image of size $M \times N$. Each sharing operation requires $(t-1)MN$ addition operations and $\frac{(t(t-1)MN)}{2}$ multiplication operations under a $(t, n)$ threshold, so their algorithmic complexity is defined as $\frac{((t+2)(t-1)MN)}{2}$. In contrast, the scheme in Hua et al. (2022) computes in groups of $(t-1)$ pixels during sharing, leading to an algorithmic complexity of $\frac{((t+2)MN)}{2}$. In the proposed scheme of this paper, each $4 \times 4$ marked pixel block only needs to compute two polynomials, each $2 \times 2$ sub-marked pixel block only needs to compute one polynomial, and each $2 \times 2$ non-sub-marked pixel block requires four polynomials to be computed. Thus, the computational complexity of this scheme depends on the texture complexity of the secret image, and not all regions use the same polynomial for sharing, resulting in lower complexity compared to other schemes.

Furthermore, this scheme effectively combines texture information to optimize the data embedding process, which not only saves space but also enhances embedding efficiency and data confidentiality. This combination allows for better implementation of data hiding and information security in practical applications, especially in scenarios involving high-resolution images or requiring the storage of large amounts of additional data.

Therefore, the proposed scheme based on generating differences using texture information not only has significant advantages in data embedding space but also demonstrates superiority in computational complexity, making it suitable for use in visual information systems that require high security and efficient computation.

## 5 Conclusion

This paper proposes a multi-party reversible data hiding algorithm based on texture-guided hierarchical quantization coding for sharing medical images. Firstly, a framework for secure multi-party storage and access of medical system information is proposed; the main data authorizers are the patients and hospitals. By implementing role-based access to information, the secure storage and access of medical images within the system are achieved. We also propose a new coding method called TGHQC, which uses the texture features of pixel blocks to adaptively generate difference information and subsequently perform hierarchical encoding. Finally, A dual embedding mechanism is designed,This embedding method significantly increases embedding capacity while ensuring information security.

**Data Availability** All data included in this study are available upon request by contact with the corresponding author.

## Declarations

**Conflict of interest** The authors have no relevant financial or non-financial interests to disclose. The authors have no Conflict of interest to declare that are relevant to the content of this article. All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript. The authors have no financial or proprietary interests in any material discussed in this article.

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